

Year 2 Quantum Mechanics - Example Sheet 2

1. The normalised ground state wavefunction for a particle confined in an infinitely deep potential well between $-a < x < a$ is given by:

$$\psi(x) = \frac{1}{\sqrt{a}} \cos(\pi x/2a).$$

Calculate the expectation values $\langle x \rangle$, $\langle p_x \rangle$, $\langle x^2 \rangle$ and $\langle p_x^2 \rangle$.

Also find $\sqrt{\langle x^2 \rangle \langle p_x^2 \rangle}$ and comment on its relevance to the Heisenberg Uncertainty Principle.

2. A proton with velocity $c/10$ moves from a region of zero potential to a region of potential energy 10 MeV. The penetration depth is the distance over which the probability reduces by a factor of e . Calculate the penetration depth.
3. A beam of electrons with energy 2 eV is incident on a barrier of height 4 eV and thickness 10^{-9} m. What is the transmission coefficient?
4. A particle in an infinite square well extending from $x = 0$ to $x = L$ has the wavefunction:

$$\Psi(x, t) = A \left(2 \sin(\pi x/L) e^{-iE_1 t/\hbar} + \sin(2\pi x/L) e^{-iE_2 t/\hbar} \right),$$

where $E_n = n^2 \pi^2 \hbar^2 / 2mL^2$.

- a) Find the value of the normalisation constant A (set $t = 0$ for simplicity).
 - b) If a measurement of the energy is made, what two values may be obtained and what is the probability of each value?
 - c) If such energy measurements are made on many particles with the above wavefunction, what will be the mean value of the energies obtained?
 - d) Calculate $\langle E \rangle$ for this wavefunction and verify that the result agrees with c).
5. A particle is confined in a one-dimensional simple harmonic oscillator potential $V(x) = \frac{1}{2}Cx^2$. The eigenfunction describing the ground state is given by:

$$\psi(x) = \frac{1}{a\sqrt{2\pi}} e^{-x^2/2a^2},$$

where $a^4 = \hbar^2/mC$.

- a) Show that this width parameter a is equal to the maximum displacement of a classical simple harmonic oscillator of energy $\hbar\omega/2$. Plot $|\psi(x)|^2$ and comment on the result.
- b) Verify that the expression $(1 + bx^2)e^{-x^2/2a^2}$ satisfies the Schrodinger equation for the simple harmonic oscillator with energy $E = 5\hbar\omega/2$ provided a is as given above. What is the required value of b ?